SIEMENS

Installation Instructions
Model ADBH-11

Audible Base

INTRODUCTION

The Model ADBH-11 Audible Base from Siemens Industry, Inc., consists of a standard Series 11 base combined with an audible device. All field wiring terminates at two four-position terminal blocks located on the back of the unit. The ADBH-11 is used with the polarity insensitive HFP-11, HFPT-11, HFPO-11, SFP-11*, SFPT-11* and SFPO-11* detectors. The ADBH-11 operates as described below (Refer to Figures 1, 2, 3 and 4).

The ADBH-11 can generate either steady or temporal tone with the insertion of a jumper at the terminal block. Refer to Tone Selection on page 7.

APPLICATION NOTE

The ADBH-11 is a UL268 listed smoke detector supplementary device which combines detector base and audible device functions. When used in conjunction with listed, compatible Siemens equipment, the ADBH-11 may be used in lieu of single/multiple station smoke alarms to achieve enhanced, system-level functionality. Typical applications are sleeping rooms in hotels, guest suites, dormitories, day-care, and residential board and care facilities as designated in NFPA 72.

*SFP-11, SFPT-11 and SFPO-11 detectors are approved for use in Canada only.

P/N 315-033210-7

Siemens Industry, Inc.
Building Technologies Division

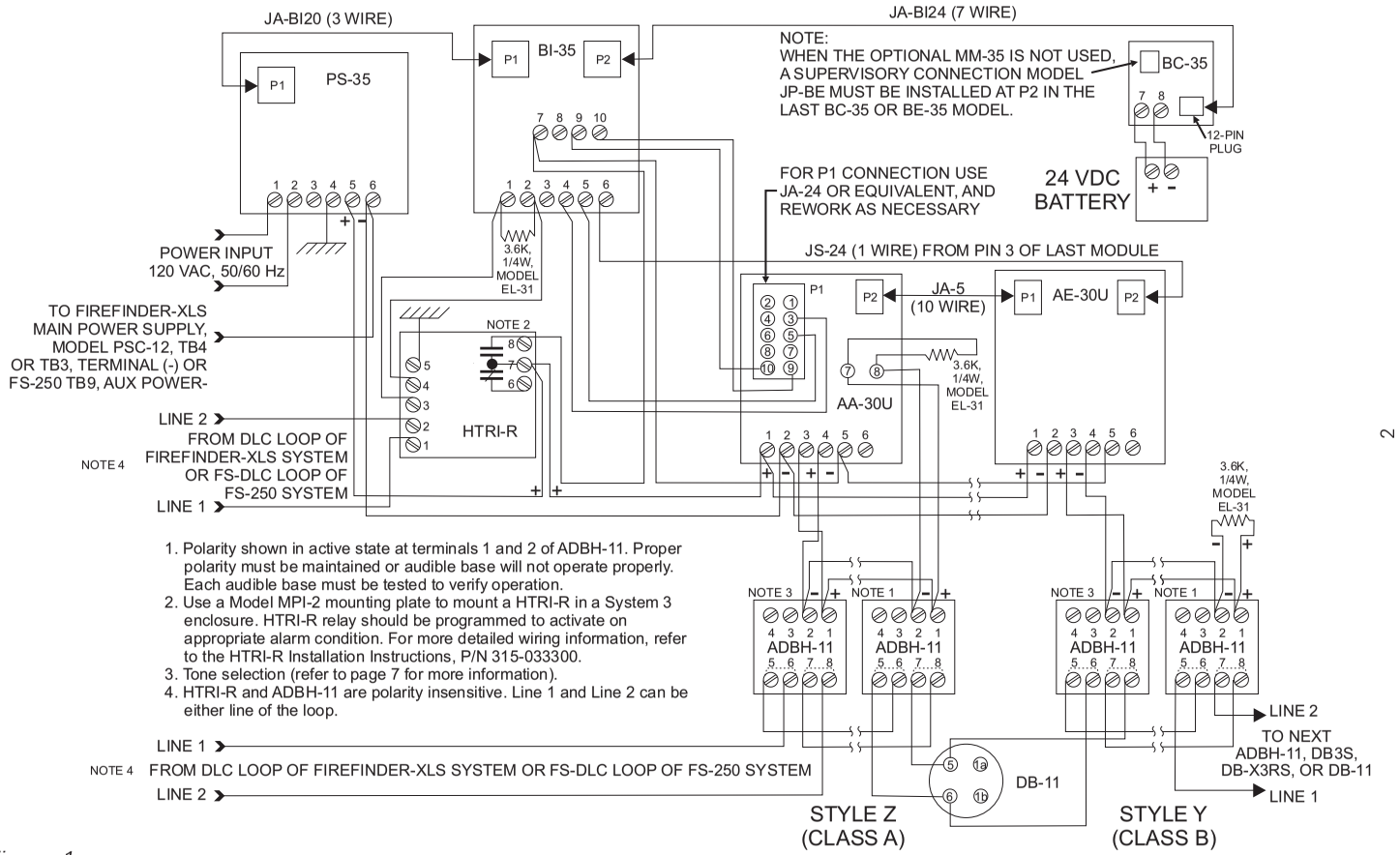
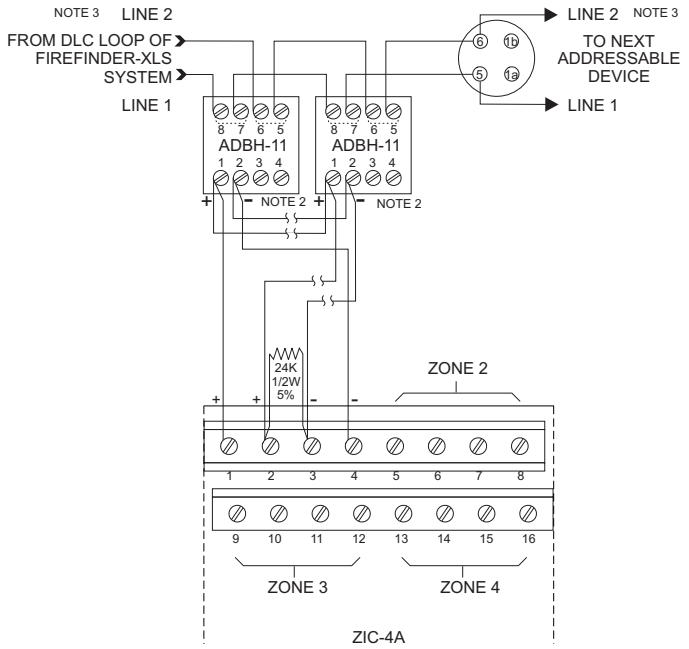


Figure 1
Installation/Wiring Diagram for ADBH-11 Powered by Auxiliary Power Supply with Battery Backup



1. Polarity shown in active state at terminals 1 and 2 of ADBH-11 and 1-4 of ZIC-4A. Proper polarity must be maintained or audible base will not operate properly. Each audible base must be tested to verify operation.
2. Tone selection (refer to page 7 for more information).
3. ADBH-11 terminals 5-8 are polarity insensitive. Line 1 and Line 2 can be either line of the loop.
4. ADBH-11 shown Style Z (Class A). It may also be wired as Style Y (Class B).
5. ZIC-4A Notification Applicance Circuits provide 4 outputs that can be configured for Class A. or Class B. Refer to the ZIC-4A Installation Instructions, P/N 315-033050.
6. EOL resistor, 24k ohms, ½ watt, 5%, P/N 140-820405.

Figure 2
Installation/Wiring Diagram for ADBH-11 Audibles Powered by ZIC-4A (FireFinder-XLS System)

ELECTRICAL RATINGS AND CONSTRAINTS

Voltage Rating: 17-31 VDC

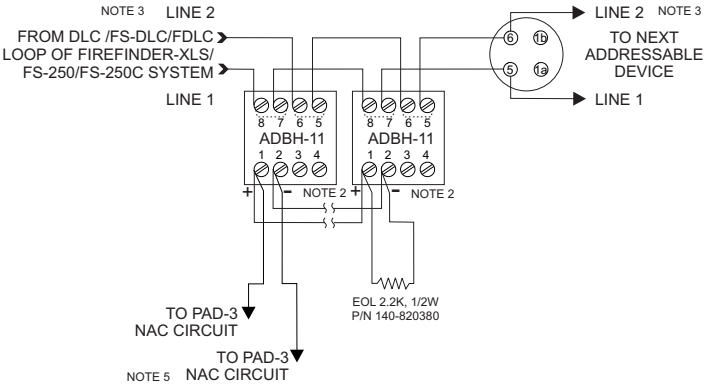
Current: Supervisory – 0A

Alarm – 24mA at 25 VDC

Minimum Sound Pressure Level: 78dB

Constraints

1. ADBH-11 is energized by an auxiliary power supply such as PS-35 with AE-30U (System 3 power supply), PAD-3 (Auxiliary power supply), ZIC-4A (FireFinder-XLS zone



1. NAC circuits polarity shown in active state. Proper polarity must be maintained or audible base will not operate properly. Each audible base must be tested to verify operation.
2. Tone selection (refer to page 7 for more information).
3. ADBH-11 is polarity insensitive. Line 1 and Line 2 can be either line of the loop.
4. ADBH-11 shown Style Z (Class A). It may also be wired as Style Y (Class B).
5. Max. Current 3.0 AMPS. Refer to PAD-3 Installation Instructions, P/N 315-099082.

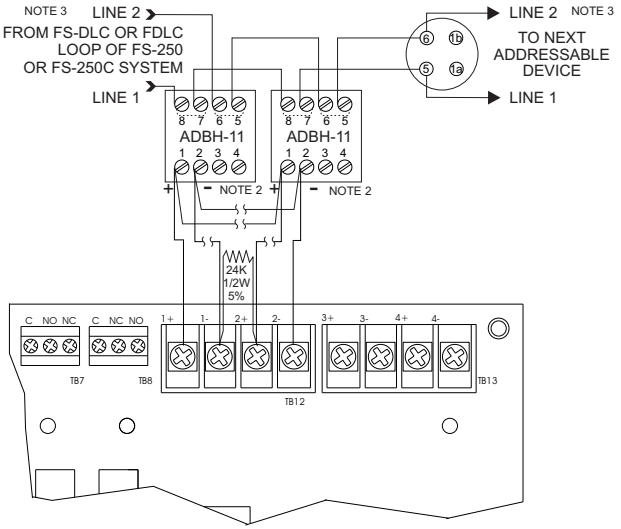
Figure 3
Installation/Wiring Diagram for ADBH-11 Audibles Powered by PAD-3

indicating card) or NAC circuits on the FS-250 or FS-250C main board. Refer to Figures 1, 2, 3 and 4. Also, any UL listed 24VDC auxiliary power supply/notification appliance circuit (listed for fire protection signaling use) can be used for ADBH-11.

2. The maximum number of ADBH-11s is related to the maximum current rating and the power line resistance of the auxiliary power supplies applied on the ADBH-11. Refer to the related installation instructions of the auxiliary power supply.

3. The maximum loop line resistance cannot exceed the maximum line resistance of the DLC/FS-DLC/FDLC driver. Refer to the DLC installation instructions, P/N 315-033090, the FS-250 Operation, Installation and Maintenance Manual, P/N 315-049353, or the FS-250C Operation, Installation and Maintenance Manual, P/N 315-049589C, as applicable.

4. T-tapping of the notification appliance circuit is not allowed.



1. Polarity shown in active state at terminals 1 and 2 of ADBH-11 and 1+, 1-, 2+ and 2- of FS-250/FS-250C NAC cirucits. Proper polarity must be maintained or audible base will not operate properly. Each audible base must be tested to verify operation.
2. Tone selection (refer to page 7 for more information).
3. ADBH-11 terminals 5-8 are polarity insensitive. Line 1 and Line 2 can be either line of the loop.
4. ADBH-11 shown Style Z (Class A). It may also be wired as Style Y (Class B).
5. FS-250/FS-250C Notification Applicance Circuits (TB12 and TB13) provide 4 outputs that can be configured for Class B or two outputs that can be configured for Class A. Refer to the FS-250 Operation, Installation and Maintenance Manual, P/N 315-049353, or the FS-250C Operation, Installation and Maintenance Manual, P/N 315-049589C, as applicable, to set jumpers JP1-4 for Class A or Class B operation.
6. EOL resistor, 24k ohms, ½ watt, 5%, P/N 140-820405.

Figure 4
Installation/Wiring Diagram for ADBH-11 Audibles Powered by NAC Circuits (FS-250/FS-250C Systems)

5. Connect terminal 6 of the PS-35 (Figure 1) to the minus DC side of the FireFinder-XLS main power supply or TB9 Aux Power minus of the FS-250 for proper ground fault supervision.

NOTE

For additional information on the modules used in an individual system, refer to the Installation Instructions for that particular module or contact your authorized Siemens Industry, Inc. representative.

DETECTOR PLACEMENT

The HFP-11, HFPT-11, HFPO-11, SFP-11, SFPT-11 and SFPO-11 detectors can be used with the ADBH-11 Detector Base. Refer to the HFP-11/HFPT-11 Installation Instructions, P/N 315-033290, the HFPO-11 Installation Instructions, P/N 315-034800, or the SFP-11, SFPT-11 and

SFPO-11 Installation Instruction, P/N 315-033290C, as applicable, for further information.

Even though no specific spacings are allocated to the detectors used with this base, use 30 foot center spacing (900 sq ft), as referred to in NFPA Standard 72 National Fire Alarm Code and CAN/ULC-S524, as a guide or starting point, if practical, in a detector installation layout. This spacing is based on ideal conditions—smooth ceiling, no air movement, and no physical obstructions.

In all installations (except in special circumstances like computer room underfloors), locate the detector on the ceiling, a minimum of 6 inches from a side wall, or on a wall, between 4 and 6 inches from the ceiling.

If questions arise regarding detector placement, it is extremely important that the drawings provided or approved by Siemens Industry, Inc., or by its authorized distributors be followed! The detector placements shown on these drawings were chosen after a careful evaluation of all facets of protecting the area. Environmental factors such as air current, temperature, humidity, air pressure, and the nature of the fire load are carefully considered. Special consideration is given to room or area configuration and the type of ceiling (sloped or flat, smooth or beamed). Siemens Industry, Inc.'s extensive experience in the design of fire detection systems assures optimum detector placement and is reflected in these drawings. Follow sound engineering judgment made by qualified personnel.

DETECTOR WIRING

(Refer to Figure 5.) Audible base Model ADBH-11 should be interconnected as shown in the Installation/Wiring Diagrams above and wired to the specific system modules and control panels following the appropriate instructions. For operation with FireFinder-XLS, FS-250 or FS-250C System, note any limitations on the number of audible bases permitted on each notification applicance circuit, depending on the total cable resistance for each audible signal circuit. (See ELECTRICAL RATINGS section.)

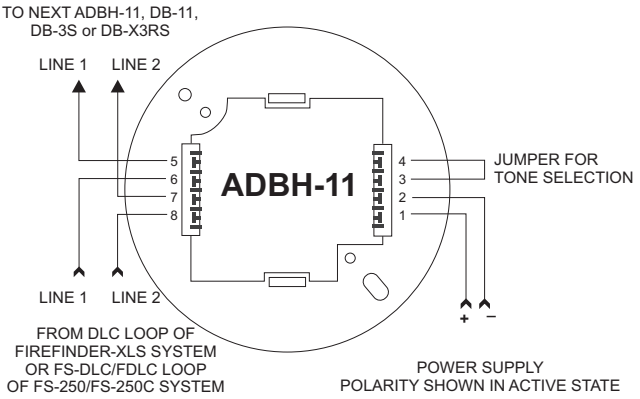


Figure 5
ADBH-11 Audible Base Terminal Blocks

Tone Selection

With audible base Model ADBH-11, the user can choose either temporal tone or steady tone. (Refer to Figure 5.)

For Temporal Tone: Terminals 3 and 4 are shorted (insert a jumper).

For Steady Tone: Terminals 3 and 4 are not connected (without a jumper).

DETECTOR MOUNTING

The detector base comes from the factory prewired. All signaling and initiating circuits are connected directly to the back of the ADBH-11 audible base using the four-position terminal blocks.

NOTE

The Audible Base Model ADBH-11 attaches only to a 4-inch square electrical box with extension.

Mounting The Base

1. Route all wires out from the outlet box.
2. When the alarm LED viewing is critical, position the LED mark on the base in the intended direction (Refer to Figure 6).
3. Make connections directly to the audible base terminals located on the back of the base. Refer to the related Installation/Wiring Diagram.

4. After all bases are installed, check loop continuity. For ADBH-11 loop continuity check, refer to the DPU or SDPU Manual, "Testing a FireFinder-XLS/FS-250 Loop" or "Testing a FS-250C Loop," as applicable.

5. If loop continuity is acceptable, proceed with detector head installation.

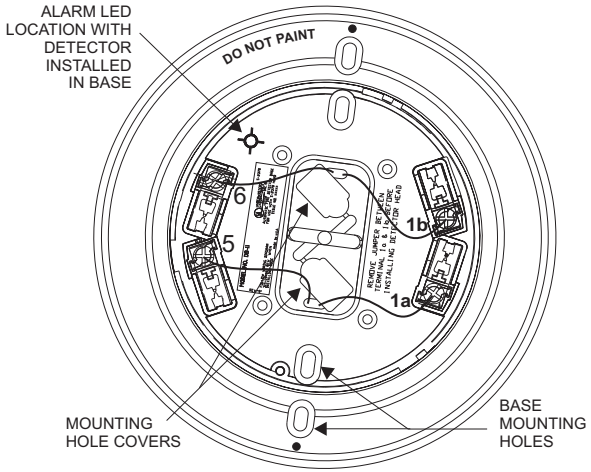


Figure 6
Positioning the Alarm LED